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Vehicle

Abstract

A vehicle may include a vehicle body including a floor panel, a battery pack located below the floor panel and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body. A support that protrudes downwardly may be disposed on a lower wall of the battery pack. The energy absorption member may include a protrusion that protrudes inwardly in the vehicle width direction toward the support. A tip end of the protrusion may include a contact surface being in contact with a side surface of the support in the vehicle width direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority to Japanese patent application No. 2022-053204, filed on Mar. 29, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The technique disclosed herein relates to a vehicle.

BACKGROUND

(3) Japanese Patent Application Publication No. 2017-226353 describes a vehicle. This vehicle includes a pair of rockers, a floor panel extending between the pair of rockers, a battery pack located below the floor panel, and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixed to one of the pair of rockers.

SUMMARY

(4) With the above structure, when side collision occurs to the vehicle, an energy absorption member (which hereafter may be referred to as an EA member) deforms, by which collision energy is absorbed. At this time, if an inner sidewall of the EA member facing the battery pack is pressed against a sidewall of the battery pack, relatively large collision load may be applied on the sidewall of the battery pack.

(5) In light of the above situation, the present disclosure provides a technique which may reduce collision load transmitted from an EA member to a sidewall of a battery pack.

(6) The technique disclosed herein is implemented as a vehicle. This vehicle may comprise: a vehicle body comprising a floor panel; a battery pack located below the floor panel; and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body. A support that protrudes downwardly may be disposed on a lower wall of the battery pack. The energy absorption member may comprise a protrusion that protrudes inwardly in the vehicle width direction toward the support. A tip end of the protrusion may comprise a contact surface being in contact with a side surface of the support in the vehicle width direction.

(7) With the above structure, when side collision occurs to the vehicle, the EA member deforms, by which collision energy is absorbed. The EA member comprises the protrusion that protrudes inwardly in the vehicle width direction toward the support disposed on the lower wall of the battery pack, and the tip end of the protrusion comprises the contact surface being in contact with the side surface of the support. Thus, when side collision occurs to the vehicle, collision load applied on the EA member is transmitted to the support disposed on the lower wall of the battery pack via the protrusion. Consequently, collision load transmitted to the sidewall of the battery pack from the EA member can be reduced.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 schematically illustrates a configuration of a vehicle **10** of Embodiment 1.

(2) FIG. 2 is a view explaining positional relationships between a battery pack **20**, a pair of EA members **30**, **32** and a pair of rockers **24**, **26**.

(3) FIG. 3 is a cross-sectional view along a line of FIG. 2.

(4) FIG. 4A illustrates an example of deformation of the EA member **30** at time when side collision occurs to the vehicle **10**; FIG. 4B illustrates an example of deformation of the EA member at a subsequent time when side collision occurs to the vehicle; and FIG. 4C illustrates an example of deformation of the EA member at another subsequent time when side collision occurs to the vehicle.

(5) FIG. 5 is an enlarged view of an enclosed area V in FIG. 3.

(6) FIG. 6 is a view to explain a variant of shapes of a contact surface **35a** of a protrusion **35** and a side surface **20c** of a support **20b**.

(7) FIG. 7 is a view to explain a variant of shapes of the contact surface **35a** of the protrusion **35** and the side surface **20c** of the support **20b**.

(8) FIG. 8 is a view to explain a variant of shapes of the contact surface 35a of the protrusion 35 and the side surface 20c of the support 20b.

(9) FIG. 9 is a view to explain a variant of shapes of the contact surface 35a of the protrusion 35 and the side surface 20c of the support 20b.

DETAILED DESCRIPTION

(10) In an embodiment of the technique disclosed herein, a dimension of the contact surface of the protrusion in a vehicle height direction may be equal to or greater than a dimension of the side surface of the support in the vehicle height direction. According to such a configuration, when side collision occurs to the vehicle, it is possible to avoid or suppress the protrusion of the EA member being locally pressed against the side surface of the support disposed on the battery pack. Consequently, it is possible to avoid or suppress collision load transmitted to the support being locally concentrated.

(11) Alternatively to the aforementioned embodiment, a dimension of the contact surface of the protrusion in a vehicle height direction may be greater than a dimension of the side surface of the support in the vehicle height direction. As another embodiment, however, the dimension of the contact surface of the protrusion in the vehicle height direction may be equal to or smaller than the dimension of the side surface of the support in the vehicle height direction.

(12) In an embodiment of the technique disclosed herein, at least a part of the protrusion may be located below the battery pack and extend along the lower wall of the battery pack. According to such a configuration, the dimension of a portion of the EA member in the vehicle width direction which is advantageous to absorb collision energy can be increased by the dimension of the protrusion in the vehicle width direction located below the battery pack. Consequently, when side collision occurs to the vehicle, the EA member deforms, by which collision energy to be absorbed can be increased.

(13) In the aforementioned embodiment, the protrusion may be fixed to the lower wall of the battery pack with a bolt. According to such a configuration, the protrusion can firmly be fixed to the battery pack. Thus, when the EA member is compressed and deformed in the vehicle width direction toward to the sidewall of the protrusion, it is possible to avoid or suppress the protrusion being unintentionally separated away from the battery pack. As another embodiment, however, the protrusion may be fixed to the lower wall of the battery pack with, for example, an adhesive instead of the bolt.

(14) In an embodiment of the technique disclosed herein, the protrusion may have a plate-like shape extending in the vehicle width direction. In this case, a thickness of the protrusion may be greater than a thickness of any one of walls that constitutes a rest of the energy absorption member. According to such a configuration, it is possible to make rigidity of the protrusion having the plate-like shape greater than rigidity of any one of the walls that constitute the rest of the EA member except the protrusion. Consequently, it is possible to make the protrusion less susceptible to deformation as compared to any one of the walls that constitute the rest of the EA member when the EA member is compressed and deformed in the vehicle width direction toward the battery pack. Thus, the collision energy generated in side collision occurring to the vehicle is primarily absorbed by deformation of a part of or an entirety of the rest of the EA member except the protrusion.

(15) In an embodiment of the technique disclosed herein, in at least a part of the protrusion adjacent to the contact surface, a thickness of the protrusion may increase toward the contact surface. According to such a configuration, it is possible to make a minimum ground clearance of the protrusion (i.e., a distance from the ground surface to the lowest surface of the protrusion in the vehicle height direction) relatively great while rigidity of the contact surface of the protrusion is maintained.

(16) In an embodiment of the technique disclosed herein, the support may be formed on a member that is attached to the battery pack. As another embodiment, however, the support may be formed integrally with the battery pack.

(17) In an embodiment of the technique disclosed herein, the contact surface of the protrusion and the side surface of the support may have uneven shapes that engage each other. According to such a configuration, it is possible to increase an area where the contact surface of the protrusion is in contact with the side surface of the support. In particular, when the uneven shapes are defined along the vehicle height direction, it is possible to suppress the contact surface of the protrusion moving away from the side surface of the support when side collision occurs to the vehicle. Consequently, when side collision occurs to the vehicle, collision load applied on the EA member is effectively transmitted to the support disposed on the lower wall of the battery pack via the protrusion.

(18) The simple recitations such as “forward”, “rearward”, “a longitudinal direction” herein respectively refer to forward, rearward and a longitudinal direction with respect to a vehicle. Similarly, the simple recitations such as “leftward”, “rightward” and “a vehicle width direction” respectively refer to leftward, rightward and a vehicle width direction with respect to the vehicle and the simple recitations such as “upward”, “downward” and “a height direction” refer to upward, downward and a height direction with respect to the vehicle. The vehicle width direction of the vehicle is also a left-right direction of the vehicle, and it may be referred to as “left-right direction” herein. For example, when the vehicle is placed on a horizontal surface, the height direction of the vehicle coincides with the vertical direction. The vehicle width direction of the vehicle is a direction parallel to the horizontal surface and parallel to an axle of the vehicle, and the longitudinal direction of the vehicle is a direction parallel to the horizontal surface and perpendicular to the axle of the vehicle.

(19) Representative, non-limiting examples of the present disclosure will now be described in further detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing aspects of the present teachings and is not intended to limit the scope of the present disclosure. Furthermore, each of the additional features and teachings disclosed below may be utilized separately or in conjunction with other features and teachings to provide improved vehicles, as well as methods for using and manufacturing the same.

(20) Moreover, combinations of features and steps disclosed in the following detailed description may not be necessary to practice the present disclosure in the broadest sense, and are instead taught merely to particularly describe representative examples of the present disclosure. Furthermore, various features of the above-described and below-described representative examples, as well as the various independent and dependent claims, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.

(21) All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter.

EMBODIMENT

(22) With reference to drawings, a vehicle **10** of an embodiment will be described. The vehicle **10** of the present embodiment is classified as an electric vehicle including a motor **16** driving wheels **14f**, **14r**, and is typically an electric vehicle which travels on roads (a so-called automobile). However, a part of or all the technique described in the present embodiment can similarly be applied to an electric vehicle which travels on tracks. The vehicle **10** is not limited to a vehicle operated and driven by a user, but may be a vehicle remotely operated by external device(s) or a vehicle which travels autonomously.

(23) Here, a direction FR in the drawings indicates a front side in a longitudinal direction (or a front-rear direction) of the vehicle **10**, and a direction RR indicates a rear side in the longitudinal direction of the vehicle **10**. A direction LH indicates a left side in a vehicle width direction (or a left-right direction) of the vehicle **10** and a direction RH indicates a right side in the vehicle width direction of the vehicle **10**. A direction UP indicates an upside in a height direction (or an up-down direction) of the vehicle **10**, and a direction DW indicates a downside in the height direction of the vehicle **10**.

(24) As illustrated in FIG. **1**, the vehicle **10** includes a vehicle body **12** and the plurality of wheels **14f**, **14r**. The vehicle body **12** includes a cabin **12c** which is a space to accommodate occupants. The plurality of wheels **14f**, **14r** is rotatably attached to the vehicle body **12**. The plurality of wheels **14f**, **14r** includes a pair of front wheels **14f** positioned at a front portion of the vehicle body **12** and a pair of rear wheels **14r** positioned at a rear portion of the vehicle body **12**. The pair of front wheels **14f** is disposed coaxially to each other, and the pair of rear wheels **14r** is also disposed coaxially to each other. The number of wheels **14f**, **14r** is not limited to four. The vehicle body **12** is constituted of, but not particularly limited to, metal, such as steel or aluminum alloy.

(25) As illustrated in FIG. **1**, the vehicle **10** further includes a motor **16**, an electric control unit **18** and a battery pack **20**. The motor **16** is a traction motor that drives the pair of rear wheels **14r** and is connected to the pair of rear wheels **14r**. The battery pack **20** is an electric device configured to supply power to the motor **16** and is electrically connected to the motor **16** via the electric control unit **18**. The battery pack **20** houses a plurality of secondary battery cells and is rechargeable with external power or regenerative power of the motor **16**. The battery pack **20** is located below a floor panel **22** and is disposed along the floor panel **22**. Although this is merely an example, as illustrated in FIGS. **2** and **3**, the battery pack **20** of the present embodiment includes brackets **20a** protruding outwardly in the vehicle width direction. A support **20b** that protrudes downwardly is disposed on a lower wall **20d** of the battery pack **20**. The support **20b** is for example a lower cover to protect the lower wall **20d** of the battery pack **20**, and is attached to the lower wall **20d** of the battery pack **20** as a member separate from the battery pack **20**. The electric control unit **18** houses a DC-DC converter and/or inverter and is configured to control driving power supplied from the battery pack **20** to the motor **16** and regenerative power supplied from the motor **16** to the battery pack **20** depending on, for example, driving operations by the user. It should be noted that it is not necessary for the support **20b** to be formed on a member that is attached to the battery pack **20**, and the support **20b** may integrally be formed with the battery pack **20**.

(26) Wheel(s) to be driven by the motor **16** are not limited to the pair of rear wheels **14r**, and the motor **16** simply needs to be configured to drive at least one of the plurality of wheels **14f**, **14r**. The vehicle **10** may further include another prime mover such as an engine instead of or as an alternative to the motor **16**. The vehicle **10** may include another power device such as fuel cell unit(s) or solar panel(s) instead of or as an alternative to the battery pack **20**. The vehicle **10** is not limited to an electric automobile described herein but may be a hybrid vehicle, fuel-cell vehicle, a solar car, or the like.

(27) As illustrated in FIGS. **2** and **3**, the vehicle body **12** includes the floor panel **22**, a pair of rockers **24**, **26** and a floor crossbeam **28**. The floor panel **22** is a plate-like member constituting a bottom surface of the cabin **12c**. Each of the rockers **24**, **26** is a hollow member having a constant cross-section along the vehicle longitudinal direction, and constitutes a frame of the vehicle body **12**. The pair of rockers **24**, **26** includes a left rocker **24** and a right rocker **26**. The left rocker **24** is positioned at a left edge **22a** of the floor panel **22** and extends in the front-rear direction on the outer side of the floor panel **22** in the vehicle width direction. The pair of rockers **24** is arranged symmetrically to each other with respect to the vehicle width direction. Thus, the right rocker **26** is positioned at a right edge of the floor panel **22** and extends in the front-rear direction on the outer side of the floor panel **22** in the vehicle width direction.

(28) Although not particularly limited to this configuration, the left rocker **24** includes a left rocker

inner panel **24a** positioned on the inner side in the width direction and a left rocker outer panel **24b** positioned on the outer side in the width direction. The left rocker inner panel **24a** and the left rocker outer panel **24b** are joined to each other at the respective top and bottom edges, and a closed space extending in the front-rear direction is defined inside the left rocker **24**. Although not particularly limited to this configuration as well, the right rocker **26** includes a right rocker inner panel **26a** positioned on the inner side in the width direction and a right rocker outer panel **26b** positioned on the outer side in the width direction. The right rocker inner panel **26a** and the right rocker outer panel **26b** are joined to each other at the respective top and bottom edges, and a closed space extending in the front-rear direction is defined inside the right rocker **26**. The floor panel **22** extends between the pair of rockers **24, 26** and is joined to the left rocker inner panel **24a** at its left edge **22a** and joined to the right rocker inner panel **26a** at its right edge. The rockers **24, 26** may be constituted of three or more panels, not limited only to the rocker inner panels **24a, 26a** and the rocker outer panels **24b, 26b**.

(29) The floor crossbeam **28** is a hollow member having a constant cross section along the vehicle longitudinal direction and constitutes a part of the frame of the vehicle body **12**. The floor crossbeam **28** is positioned on the floor panel **22** and extends between the pair of rockers **24, 26** in the vehicle width direction. Although the illustration is omitted, a plurality of floor crossbeams **28** is disposed between the pair of rockers **24, 26**.

(30) As illustrated in FIGS. **2, 3**, the vehicle **10** further includes a pair of energy absorption members **30, 32** (hereafter referred to as EA members **30, 32**). Each of the pair of EA members **30, 32** is a hollow member extending along the vehicle longitudinal direction and having a constant cross section along the vehicle longitudinal direction. The pair of EA members **30, 32** includes a left EA member **30** and a right EA member **32**. The pair of EA members **30, 32** is arranged symmetrically to each other with respect to the vehicle width direction. The left EA member **30** is located at a left portion of the vehicle **10** and outwardly from the battery pack **20** in the vehicle width direction. The right EA member **32** is located at a right portion of the vehicle **10** and outwardly from the battery pack **20** in the vehicle width direction. One of the EA members **30, 32** is compressed and deformed when side collision occurs to the vehicle **10**, by which it absorbs collision energy. The pair of EA members **30, 32** is constituted of metal such as aluminum. However, a material constituting the pair of EA members **30, 32** is not particularly limited.

(31) As described above, the pair of EA members **30, 32** is arranged symmetrically to each other with respect to the vehicle width direction. Structures of the left EA member **30** and its relevant members (e.g., the battery pack **20** and the left rocker **24**) are symmetrical to structures of the right EA member **32** and its relevant members (e.g., the battery pack **20** and the right rocker **26**) with respect to the vehicle width direction. Thus, with reference to FIG. **3**, the structures of the left EA member **30** and its relevant members (e.g., the battery pack **20** and the left rocker **24**) will hereafter be described. Hereafter, the left EA member **30** will simply be referred to as the EA member **30**.

(32) As illustrated in FIG. **3**, the EA member **30** includes an inner end portion **34**, a first middle portion **36**, a second middle portion **38** and an outer end portion **40**. The inner end portion **34** is at the innermost position in the vehicle width direction and faces the battery pack **20**. The outer end portion **40** is at the outermost position in the vehicle width direction. The first middle portion **36** and the second middle portion **38** are positioned between the inner end portion **34** and the outer end portion **40** in the vehicle width direction. The first middle portion **36** is adjacent to the inner end portion **34** and the second middle portion **38** is adjacent to the outer end portion **40**. In other words, the inner end portion **34**, the first middle portion **36**, the second middle portion **38** and the outer end portion **40** are arranged in this order from inside to outside in the vehicle width direction.

(33) Although not particularly limited to this configuration, the inner end portion **34** includes an inner sidewall **34a**, a vertical wall **34b** and a lower wall **34c**. The inner sidewall **34a** faces the battery pack **20** and inclines downwardly in the vehicle height direction as it extends inwardly in the vehicle width direction. The vertical wall **34b** extends along the vehicle height direction and the

lower wall **34c** extends along the vehicle width direction. Thus, a space having a constant cross-section in the vehicle longitudinal direction is defined by the inner sidewall **34a**, the vertical wall **34b** and the lower wall **34c**.

(34) As illustrated in FIG. 3, the vehicle **10** further includes first collars **42** and first bolts **44**. The inner sidewall **34a** included in the inner end portion **34** of the EA member **30** comprises openings through which the first collars **42** extend. The first collars **42** are disposed such that the first collars **42** extend through the openings in the inner sidewall **34a** in the vehicle height direction. Although this is an example, the first collars **42** are welded to the inner sidewall **34a** of the EA member **30**. The upper end of each first collar **42** is in contact with corresponding one of the brackets **20a** of the battery pack **20** and the lower end of each first collar **42** is in contact with the lower wall **34c** of the inner end portion **34**. The first bolts **44** extend through the first collars **42** and fasten the EA member **30** to the brackets **20a** of the battery pack **20**. Thus, the EA member is fixed to the brackets **20a** of the battery pack **20** at the inner end portion **34**. An inner diameter **D1** of the first collars **42** is greater than a nominal diameter **D2** of the first bolts **44**. For example, the inner diameter **D1** of the first collars **42** is at least twice the nominal diameter **D2** of the first bolts **44**. As another embodiment, the brackets **20a** may not be disposed on the battery pack **20**, and the EA member **30** may directly be fastened to the battery pack **20** by the first bolts **44**.

(35) As illustrated in FIG. 3, the vehicle **10** further includes second collars **46** and second bolts **48**. Although not particularly limited to this configuration, an upper wall **38a** of the second middle portion **38** of the EA member **30** also includes openings through which the second collars **46** extend. The second collars **46** are disposed such that the second collars **46** extend through the openings in the upper wall **38a** of the second middle portion **38** in the vehicle height direction. Although this is an example, the second collars **46** are welded to the upper wall **38a** of the second middle portion **38**. The upper ends of the second collars **46** are in contact with the left rocker inner panel **24a** of the left rocker **24** and the lower ends of the second collars **46** are in contact with the lower wall **38b** of the second middle portion **38**. The second bolts **48** extend through the second collars **46** and fasten the EA member **30** to the left rocker **24**. Thus, the EA member **30** is fixed to the left rocker **24** of the vehicle body **12** at the second middle portion **38**. As described above, the EA member **30** is fixed to the brackets **20a** of the battery pack **20** at the inner end portion **34**, thus the EA member **30** fixes the battery pack **20** to the vehicle body **12**. Although not particularly limited to this configuration, the inner diameter of the second collars **46** is greater than the nominal diameter of the second bolts **48**.

(36) It should be noted that, the position where the EA member **30** is fixed to the bracket **20a** of the battery pack **20** does not need to be the inner end portion **34**. The position where the EA member **30** is fixed to the bracket **20a** of the battery pack **20** can suitably be changed depending on, for example, a shape of the EA member **30** and/or a positional relationship between the EA member **30** and the battery pack **20**. Similarly, the position where the EA member **30** is fixed to the left rocker **24** of the vehicle body **12** does not need to be the second middle portion **38**. The position where the EA member **30** is fixed to the left rocker **24** of the vehicle body **12** can suitably be changed depending on, for example, the shape of the EA member **30** and a positional relationship between the EA member **30** and the left rocker **24**.

(37) As illustrated in FIG. 3, the EA member **30** further includes a protrusion **35**. The protrusion **35** is connected to the lower wall **34c** and the inner sidewall **34a** of the inner end portion **34** and extends inwardly in the vehicle width direction toward the support **20b** disposed on the battery pack **20**. The protrusion **35** is located below the battery pack **20** and extends along the lower wall **20d** of the battery pack **20**. The protrusion **35** is fixed to the lower wall **20d** of the battery pack **20** with a third bolt **50**. The tip end of the protrusion **35** comprises a contact surface **35a** being in contact with a side surface **20c** of the support **20b** in the vehicle width direction. An entirety of the protrusion **35** needs not to be located below the battery pack **20**, and only at least one of the protrusion **35** has to be located below the battery pack **20**. As another embodiment, the protrusion

35 may be fixed to the lower wall **20d** of the battery pack **20** with for example an adhesive, instead of the third bolt **50**.

(38) With the aforementioned structure, as illustrated in FIGS. **4A** to **4C**, one of the EA members **30**, **32** deforms when side collision occurs to the vehicle **10**, by which collision energy is absorbed. Each of the EA members **30**, **32** includes the protrusion **35** which protrudes inwardly in the vehicle width direction toward the support **20b** disposed on the lower wall **20d** of the battery pack **20**, and the tip end of the protrusion **35** comprises the contact surface **35a** being in contact with the side surface **20c** of the support **20b**. Thus, when side collision occurs to the vehicle **10**, collision load applied on one of the EA members **30**, **32** is transmitted to the support **20b** disposed on the lower wall **20d** of the battery pack **20**. Consequently, the collision load transmitted from the one of the EA members **30**, **32** to the corresponding sidewall of the battery pack **20** can be reduced.

(39) Although not particularly limited to this configuration, as illustrated in FIG. **5**, the dimension **H1** of the contact surface **35a** of the protrusion **35** in the vehicle height direction is equal to or greater than the dimension **H2** of the side surface **20c** of the support **20b** in the vehicle height direction, and more specifically, greater than the dimension **H2** of the side surface **20c** of the support **20b** in the vehicle height direction. According to such a configuration, when side collision occurs to the vehicle **10**, it is possible to avoid or suppress the protrusion **35** of one of the EA members **30**, **32** being locally pressed against the corresponding side surface **20c** of the support **20b** disposed on the battery pack **20**. Consequently, it is possible to avoid or suppress collision load transmitted to the support **20b** being locally concentrated. As another embodiment, however, the dimension **H1** of the contact surface **35a** of the protrusion **35** in the vehicle height direction may be smaller than the dimension **H2** of the side surface **20c** of the support **20b** in the vehicle height direction.

(40) Although not particularly limited to this configuration, as illustrated in FIG. **5**, the protrusion **35** has a plate-like shape extending in the vehicle width direction. In this case, the thickness **T1** of the protrusion **35** is greater than the thickness of any one of the walls that constitute the rest of the EA member **30** (i.e., each of the thicknesses **T2**, **T3** of the walls **34a** to **34c** that constitute the inner end portion **34**, a thickness of any one of walls constituting the first middle portion **36**, a thickness of each of the walls **38a**, **38b** that constitute the second middle portion **38** and a thickness of any one of walls that constitute the outer end portion **40**). According to such a configuration, in the EA members **30**, **32**, it is possible to make rigidity of the protrusions **35** having the plate-like shape greater than rigidity of any one of the walls that constitute the rest of the EA members **30**, **32** except the protrusions **35**. Consequently, when one of the EA members **30**, **32** is compressed and deformed in the vehicle width direction toward the battery pack **20**, it is possible to make the corresponding protrusion **35** less susceptible to deformation as compared to any one of the walls that constitute the rest of the one of the EA members **30**, **32**, which is the inner end portion **34**, the first middle portion **36**, the second middle portion **38**, and the outer end portion **40**. Thus, the collision energy generated in side collision occurring to the vehicle **10** is primarily absorbed by deformation of a part of or the entirety of the rest of the one of the EA members **30**, **32** except the protrusion **35**.

(41) Although not particularly limited to this configuration, as illustrated in FIG. **5**, in at least a part of the protrusion **35** adjacent to the contact surface **35a**, the thickness of the protrusion **35** increases toward the contact surface **35a**. In other words, the thickness **T1** of the protrusion **35** changes such that it increases toward the contact surface **35a**. For example, a thickness of a part of the protrusion **35** at which the third bolt **50** is disposed can relatively be reduced and a thickness of the contact surface **35a** at the tip end of the protrusion **35** (i.e., the dimension **H1** of the contact surface **35a** in the vehicle height direction) can relatively be increased. According to such a configuration, it is possible to make the minimum ground clearance of the protrusion **35** (i.e., a distance from the ground surface to the lowest surface of the protrusion **35** in the vehicle height direction) relatively great while rigidity of the contact surface **35a** of the protrusion **35** is maintained.

(42) Alternatively to the aforementioned embodiment, as illustrated in FIGS. 6 to 9, the thickness T1 of the protrusion 35 may be constant. In this case, the thickness T1 of the protrusion 35 may be equal to the dimension H1 of the contact surface 35a of the protrusion 35 in the vehicle height direction. The contact surface 35a disposed at the tip end of the protrusion 35 and the side surface 20c of the support 20b can have various shapes. For example, as illustrated in FIGS. 7-9, the contact surface 35a disposed at the tip end of the protrusion 35 has a shape protruding inwardly in the vehicle width direction toward the side surface 20c of the support 20b. The side surface 20c of the support 20b is recessed inwardly in the vehicle width direction to receive the contact surface 35a of the protrusion 35. As described above, when the contact surface 35a of the protrusion 35 and the side surface 20c of the support 20b have uneven shapes which engage each other, it is possible to increase an area where the contact surface 35a of the protrusion 35 is in contact with the side surface 20c of the support 20b. In particular, when the uneven shapes are defined along the vehicle height direction, it is possible to suppress the contact surface 35a of the protrusion 35 moving away from the side surface 20c of the support 20b upon side collision to the vehicle 10. Consequently, when side collision occurs to the vehicle 10, collision load applied on one of the EA members 30, 32 is effectively transmitted to the support 20b disposed on the lower wall 20d of the battery pack 20 via the corresponding protrusion 35.

Claims

1. A vehicle comprising: a vehicle body comprising a floor panel; a battery pack located below the floor panel; and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body, wherein: a support that protrudes downwardly is disposed on a lower wall of the battery pack, the energy absorption member comprises a protrusion that protrudes inwardly in the vehicle width direction toward the support, a tip end of the protrusion comprises a contact surface being in contact with a side surface of the support in the vehicle width direction, and a dimension of the contact surface of the protrusion in a vehicle height direction is equal to or greater than a dimension of the side surface of the support in the vehicle height direction.
2. The vehicle according to claim 1, wherein at least a part of the protrusion is located below the battery pack and extends along the lower wall of the battery pack.
3. The vehicle according to claim 2, wherein the protrusion is fixed to the lower wall of the battery pack with a bolt.
4. The vehicle according to claim 1, wherein, in at least a part of the protrusion adjacent to the contact surface, a thickness of the protrusion increases toward the contact surface.
5. The vehicle according to claim 1, wherein the support is formed on a member that is attached to the battery pack.
6. The vehicle according to claim 1, wherein the support is formed integrally with the battery pack.
7. The vehicle according to claim 1, wherein the contact surface of the protrusion and the side surface of the support have uneven shapes that engage each other.
8. A vehicle comprising: a vehicle body comprising a floor panel; a battery pack located below the floor panel; and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body, wherein: a support that protrudes downwardly is disposed on a lower wall of the battery pack, the energy absorption member comprises a protrusion that protrudes inwardly in the vehicle width direction toward the support, a tip end of the protrusion comprises a contact surface being in contact with a side surface of the support in the vehicle width direction, and a dimension of the contact surface of the protrusion in a vehicle height direction is greater than a dimension of the side surface of the support in the vehicle height direction.
9. A vehicle comprising: a vehicle body comprising a floor panel; a battery pack located below the

floor panel; and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body, wherein: a support that protrudes downwardly is disposed on a lower wall of the battery pack, the energy absorption member comprises a protrusion that protrudes inwardly in the vehicle width direction toward the support, a tip end of the protrusion comprises a contact surface being in contact with a side surface of the support in the vehicle width direction, and at least a part of the protrusion is located below the battery pack and extends along the lower wall of the battery pack.

10. The vehicle according to claim 9, wherein the protrusion is fixed to the lower wall of the battery pack with a bolt.
